Media Engineering and Technology Faculty
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Portfolio Optimization: Asset Universe Selection Based on Investor Profiles

Bachelor Thesis

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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowlegement has been made in the text to all other material used

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30 March, 2026

Acknowledgments

Acknowledgment goes here.

Abstract

Abstract text goes here.

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Chapter 1

Introduction

1.1 Motivation

Portfolio construction is usually presented as an optimization problem over a set of available assets. Classical portfolio theory explains how capital can be distributed once that set is known, but it often leaves an earlier question outside the model: which assets should be considered in the first place? This question matters because the same candidate universe is not suitable for every investor. Before portfolio weights are assigned, the asset universe should be checked against the investor's objectives, risk tolerance, and the current market conditions.

This issue is important in markets where asset behavior changes over time. In the Egyptian financial market, risk can shift because of inflation, exchange-rate movements, monetary-policy changes, and market stress. In such conditions, treating an asset as permanently conservative or permanently aggressive can be misleading. A portfolio process that starts directly with final allocation may therefore depend on a candidate universe that is already weak for the target investor profile.

This thesis treats asset-universe selection as a separate decision stage before allocation. The goal is not to compute final portfolio weights. Instead, the thesis studies how artificial intelligence can support the selection of an investor-suitable universe before an allocation model is applied. Investor suitability is interpreted through three risk-tolerance groups. Conservative investors prioritize risk control, balanced investors seek a risk-return tradeoff, and aggressive investors accept higher risk in exchange for higher return opportunity.

The implementation studies this problem through an adaptive risk-scoring approach. A reinforcement-learning model predicts an asset-level composite realized-risk score and ranks the active universe in each decision month. The realized-risk target combines realized volatility, downside deviation, and maximum drawdown. These predictions are then used to form risk-tolerance-oriented asset universes before allocation. The contribution is therefore AI-supported asset-risk prediction for investor-suitable universe construction, not final portfolio optimization.

1.2 Problem Statement

Existing portfolio allocation methods generally assume that the investable universe has already been defined. Related AI and machine-learning preselection studies show that filtering assets before optimization is a valid modeling stage. However, these studies commonly select assets using expected return, price movement, profitability, Sharpe-like measures, efficiency, robustness, or final optimizer performance. The problem addressed in this thesis is how to select an investor-suitable asset universe before allocation when suitability is defined by risk tolerance and asset-level realized-risk behavior.

1.3 Scope and Asset Universe

The thesis focuses on a mixed Egyptian-market universe. The investable asset universe includes money-market exposure represented by 91-day Treasury bills, government bonds represented by 5-year bonds, Egyptian equity exposure represented by EGX30 and available EGX30 constituent stocks, real-estate exposure represented by a REIT series, and gold exposure represented by 24K gold in Egyptian pounds. Macroeconomic inputs, including USD exchange-rate behavior and CPI, are used only as model features. They are not treated as investable assets.

The current scope ends at pre-allocation universe selection. Final portfolio weighting, investor-facing production serving, and allocation simulation are not treated as the main implemented contribution. They are later stages that can use the selected universe produced by the proposed framework.

1.4 Research Questions and Objectives

This thesis is guided by the following research questions:

- RQ1: Can AI/ML support dynamic asset-universe selection before allocation using asset-level realized-risk prediction?
- RQ2: Do the selected universes align with conservative, balanced, and aggressive investor risk-tolerance profiles?
- RQ3: How does direct asset-risk prediction differ from existing AI/ML preselection criteria such as return prediction, price movement, Sharpe-like performance, efficiency, or optimizer quality?

The objectives of the thesis are to:

- design a pre-allocation asset-universe selection framework guided by investor risk tolerance;
- construct point-in-time asset-risk features and composite realized-risk targets;
- train and evaluate a reinforcement-learning model that predicts asset-level realized risk;
- evaluate the selected universes against full-universe, random, oracle, and heuristic baselines.

1.5 Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on portfolio optimization, AI/ML asset preselection, investor-risk personalization, reinforcement-learning methods, and the research gap addressed by this work. Later chapters present the methodology, experimental evaluation, and conclusions of the proposed framework.

Chapter 2

Literature Review

2.1 Framing the Problem

This thesis studies asset-universe selection as a stage that comes before final portfolio weighting. An optimizer can only allocate among assets that are included in the candidate set. If that set contains assets that do not match the investor's risk tolerance, then the allocation stage starts from a weak position. For this reason, this chapter reviews classical portfolio optimization, AI/ML asset preselection, investor-risk personalization, and adaptive reinforcement-learning methods.

The review is organized around one central distinction. Many existing papers show that selecting assets before portfolio optimization can improve later portfolio construction. However, a suitable asset is often defined through expected return, price movement, profitability, Sharpe-like performance, efficiency, robustness, or final optimizer quality. This thesis defines suitability differently. It studies whether assets can be selected for conservative, balanced, and aggressive investor groups by directly predicting asset-level realized risk before allocation.

2.2 Portfolio Optimization and the Fixed-Universe Assumption

A classical starting point is the mean-variance framework introduced by Markowitz [1]. The framework formalizes the tradeoff between return and risk when assigning portfolio weights, but it assumes that the investable universe is already known. This assumption is important here because it separates two decisions that are often combined in practice: choosing the candidate assets and assigning final weights.

Later allocation methods provide additional ways to represent investor views and risk objectives. The Black-Litterman model incorporates investor views into allocation decisions [2], while conditional value-at-risk optimization focuses on downside tail risk [3]. Quantile-based portfolio selection also extends the discussion beyond variance by studying distributional risk criteria [4]. These approaches are relevant because they show that risk can be modeled in several ways, but they still mainly operate after the asset universe has been defined.

Benchmarking work also shows why the universe-selection stage should be separated from the weighting stage. DeMiguel et al. [5] showed that naive diversification is a difficult benchmark for optimized portfolios to beat, while hierarchical risk parity later offered a risk-based allocation alternative that emphasizes diversification structure [6]. These methods are useful baselines for allocation, but they do not decide which assets are suitable for a specific investor before weights are computed.

2.3 AI/ML Asset Preselection Before Optimization

The closest baseline family to this thesis is AI/ML-based asset preselection before portfolio optimization. Wang et al. [7] proposed a deep-learning preselection framework where selected assets are passed to portfolio formation. This paper is directly relevant because it treats asset filtering as a meaningful stage before allocation. Its selection logic, however, is not based on direct prediction of an asset-level realized-risk target for investor risk-tolerance grouping.

Similar predict-then-select patterns appear in later studies. Ma et al. [8] connected machine-learning and deep-learning return prediction to portfolio optimization. Kaczmarek and Perez [9] used machine-learning predictions to build portfolios under rules such as mean-variance optimization, hierarchical risk parity, and naive diversification. Chaweewanchon and Chaysiri [10] used predictive stock selection before Markowitz optimization. These studies show that ML can serve as a filter before final allocation, but the filtering signal is mainly driven by return or price prediction.

Other work broadens preselection beyond simple return prediction. Mills and Anyomi [11] proposed a two-stage robust portfolio construction approach under uncertainty. Hosseinzadeh et al. [12] used data envelopment analysis for asset preselection before portfolio optimization. Abdi et al. [13] combined LSTM modeling with Sharpe-ratio maximization for prospective asset preselection. Chou and Pham [14] used ensemble learning for stock preselection with multiobjective investment optimization and investor-preference framing.

Taken together, these papers establish that preselection before allocation is a valid modeling stage. They also form the main baseline family for this thesis. The key difference is the definition of suitability. In most of this literature, a suitable asset is one expected to improve return, price prediction, efficiency, Sharpe-like performance, robustness, or optimizer quality. In this thesis, a suitable asset is one whose predicted realized-risk behavior matches the investor's risk-tolerance group before any allocation model is applied.

2.4 Investor Risk Tolerance and Personalization

The preselection literature motivates the stage studied in this thesis, but investor-specific suitability also requires work on personalization. Personalized finance advisory systems show that recommendations can change when investor information is modeled explicitly. Musto et al. [15] used case-based recommendation and diversification strategies for personalized finance advice. Although the output is not a formal asset universe, the paper supports the principle that financial recommendation should not treat all investors as identical.

Recent AI-based personalization studies provide additional support for investor-aware financial decisions. Asemi et al. [16] used an adaptive neuro-fuzzy inference system to customize investment type from investor demographics and feedback. A related model combined ANFIS, clustering, expert feedback, and multimodal neural-network pretraining for investment-type recommendation [17]. Wei and Liu [18] proposed personalized fund recommendation with dynamic utility learning. These papers connect investor information to financial recommendations, but their outputs are usually investment types or fund recommendations rather than an asset-level universe selected before allocation.

Risk tolerance is also studied directly in robo-advising and personalized portfolio design. Alsabah et al. [19] modeled how investor risk preferences can be learned from portfolio choices. Yu and Liu [20] proposed a personalized mean-CVaR portfolio optimization model for individual investment. Capponi et al. [21] studied personalized robo-advising through client interaction. Schneider and Yilmaz [22] showed that risk appetite can change stock and portfolio recommendations produced by ChatGPT. These studies justify the use of investor risk tolerance as a decision variable, but they do not usually define an explicit pre-allocation asset universe selected through direct asset-risk prediction.

2.5 Adaptive AI and Reinforcement Learning in Finance

Adaptive AI methods are relevant because financial suitability is not static. Deep reinforcement learning has been used for trading and portfolio-management tasks, including financial signal representation and trading decisions [23]. Jiang et al. [24] proposed a deep reinforcement-learning framework for portfolio management, and FinRL later provided environments and tools for automated stock trading and quantitative finance research [25].

Additional DRL portfolio studies show how adaptive policies can be evaluated in financial settings. Lucarelli and Borrotti [26] studied deep Q-learning for cryptocurrency portfolio management. Soleymani and Paquet [27] used online deep reinforcement learning with an autoencoder representation for portfolio optimization. Pinelis and Ruppert

[28] studied machine-learning portfolio allocation, while Rezaei and Nezamabadi-Pour [29] provided a taxonomy and experimental study of deep reinforcement learning in portfolio management.

Risk-aware reinforcement learning is especially relevant to the implementation direction of this thesis. Choudhary et al. [30] trained risk-adjusted deep reinforcement-learning agents using reward functions related to return, Sharpe ratio, and maximum drawdown. This supports the use of RL in risk-aware financial modeling. However, the action in such work is still usually a trading or allocation decision, not the prediction of an asset-level risk score used to form a pre-allocation universe.

The closest paper to the present thesis is the investor-specific DRL framework of Orra et al. [31]. Their pipeline uses volatility-guided grouping to form conservative, moderate, and aggressive risk profiles before DRL allocation. This is highly relevant because it connects investor risk profiles, asset grouping, and adaptive learning. The remaining difference is that the grouping is volatility-guided and stock-focused, while this thesis predicts a composite realized-risk target based on realized volatility, downside deviation, and maximum drawdown over a mixed and variable asset universe.

2.6 Synthesis and Research Gap

The reviewed literature leads to three observations. First, classical allocation methods provide strong tools for assigning portfolio weights, but they usually assume that the investable universe has already been chosen. Second, AI/ML preselection studies show that selecting assets before optimization is a valid modeling stage. Third, investor-personalization studies show that risk tolerance should affect financial recommendations. The gap appears between these streams rather than inside only one of them.

The main baseline family for this thesis is AI/ML asset preselection before portfolio optimization. These studies solve a related problem and should be treated as important comparison work. Their limitation is more specific: the selection criterion is usually return prediction, price movement, profitability, Sharpe-like performance, efficiency, robustness, or optimizer quality. Direct prediction of an asset-level risk target is mostly absent. The closest partial exception is Orra et al. [31], which uses volatility for risk-profile grouping, but it does not predict a composite realized-risk target in the same form used here.

This thesis is positioned as an investor-risk-tolerance-oriented extension of the preselection literature. It studies pre-allocation universe construction by predicting and ranking assets according to a composite `realized_risk` target before final allocation. The selected universes are then interpreted according to conservative, balanced, and aggressive investor profiles. In this way, the thesis connects asset preselection, investor risk tolerance, and adaptive AI while keeping final portfolio weighting as a later stage.

Chapter 3

Methodology

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Chapter 4

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4.4 Results Analysis and Discussion

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Chapter 5

Conclusion & Future Work

5.1 Conclusion

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5.2 Future Work

Section text.

Appendix

Appendix A

Lists

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